Lecture 6

Compact Structure and Maximal Tori

Objectives. *This lecture explains why conjugacy in a compact connected group can be studied on a torus. First, an averaged inner product splits the Lie algebra. Next, maximal tori meet every conjugacy class. Finally, a finite Weyl group records which elements of the torus are conjugate.*

By the end of this lecture you will be able to:

- Average an inner product and prove $\mathfrak{k} = \mathfrak{z}(\mathfrak{k}) \oplus [\mathfrak{k}, \mathfrak{k}]$ with $[\mathfrak{k}, \mathfrak{k}]$ semisimple.
- Prove that maximal tori are conjugate, explain the strategy of the torus theorem, and deduce that exp is surjective for compact connected groups.
- Compute $N_K(T)/T$ for $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ and match it with the algebraic Weyl group.
- Locate where these statements fail for $\mathrm{SL}(2, \mathbb{R})$.

Throughout, K is a compact Lie group with Lie algebra $\mathfrak{k}$. Its Haar probability measure dg is left and right invariant (Lecture 5). Connectedness of K is assumed only where a statement says so.

6.1 An invariant inner product and the splitting

Lemma 6.1. *There is an inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{k}$ such that for all $g \in K$ and $X, Y, Z \in \mathfrak{k}$*

$$\langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle = \langle X, Y \rangle, \qquad \langle [Z, X], Y \rangle = -\langle X, [Z, Y] \rangle. \tag{6.1}$$

Proof. Choose any inner product $\langle \cdot, \cdot \rangle_0$ on $\mathfrak{k}$ and set

$$\langle X, Y \rangle = \int_K \langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle_0 \, dg.$$

The integrand is continuous in g. So the form is well defined, symmetric and bilinear. For $X \neq 0$ the continuous function $g \mapsto \| \mathrm{Ad}(g)X \|_0^2$ is positive on the compact set K. It has a positive minimum m, so $\langle X, X \rangle \geq m > 0$. Right invariance of dg gives

$$\langle \mathrm{Ad}(h)X, \mathrm{Ad}(h)Y \rangle = \int_K \langle \mathrm{Ad}(gh)X, \mathrm{Ad}(gh)Y \rangle_0 \, dg = \langle X, Y \rangle.$$

For the second identity, recall $\mathrm{Ad}(\exp tZ) = e^{t \, \mathrm{ad} Z}$ from Lecture 2. By the first identity, the function $t \mapsto \langle e^{t \, \mathrm{ad} Z} X, e^{t \, \mathrm{ad} Z} Y \rangle$ is constant. By the product rule, its derivative at $t = 0$ is

$$0 = \langle [Z, X], Y \rangle + \langle X, [Z, Y] \rangle.$$

∎

In other words, every $\operatorname{ad} Z$ is skew-symmetric for $\langle \cdot, \cdot \rangle$. From now on we fix such an inner product.

Proposition 6.2. *Let $\mathfrak{z} = \mathfrak{z}(\mathfrak{k})$ be the centre and $\mathfrak{s} = [\mathfrak{k}, \mathfrak{k}]$.*

(a) $\mathfrak{s}^{\perp} = \mathfrak{z}$. Hence $\mathfrak{k} = \mathfrak{z} \oplus \mathfrak{s}$, an orthogonal sum of ideals.

(b) The Killing form B of $\mathfrak{s}$ is negative definite. Hence $\mathfrak{s}$ is semisimple.

Proof. (a) For $Z, X, Y \in \mathfrak{k}$, the skew-symmetry identity in (6.1) gives

$$\langle Z, [X, Y] \rangle = -\langle [X, Z], Y \rangle.$$

So $Z \perp \mathfrak{s}$ exactly when $[X, Z] \perp Y$ for all X, Y. This means $[X, Z] = 0$ for all X, that is, $Z \in \mathfrak{z}$. Both $\mathfrak{z}$ and $\mathfrak{s}$ are ideals. Since $\mathfrak{k}$ is finite-dimensional, $\mathfrak{k} = \mathfrak{s}^{\perp} \oplus \mathfrak{s} = \mathfrak{z} \oplus \mathfrak{s}$.

(b) First, $\mathfrak{z}(\mathfrak{s}) = 0$. An element of $\mathfrak{z}(\mathfrak{s})$ commutes with $\mathfrak{s}$ and with $\mathfrak{z}$, hence with $\mathfrak{k}$ by the splitting in (a). So it lies in $\mathfrak{z} \cap \mathfrak{s} = 0$.

Now let $X \in \mathfrak{s}$ and $A = \operatorname{ad} X|_{\mathfrak{s}}$. Since $\mathfrak{s}$ is an ideal, A maps $\mathfrak{s}$ to itself. The map A is skew-symmetric, since every $\operatorname{ad} X$ is. So in an orthonormal basis $e_1, \dots, e_m$ of $\mathfrak{s}$ the matrix of A satisfies $A^{\mathsf{T}} = -A$. Hence

$$B(X, X) = \operatorname{tr}(A^2) = -\operatorname{tr}(A^{\mathsf{T}} A) = -\sum_{i=1}^{m} \|Ae_i\|^2 \le 0.$$

Equality forces $A = 0$, so $X \in \mathfrak{z}(\mathfrak{s}) = 0$. Thus B is negative definite.

The complexification of B is the Killing form of $\mathfrak{s}_{\mathbb{C}}$, and it is nondegenerate. By the complex Cartan criterion from the Lie Algebras course, $\mathfrak{s}_{\mathbb{C}}$ is semisimple. A nonzero solvable real ideal of $\mathfrak{s}$ would complexify to a nonzero solvable ideal of $\mathfrak{s}_{\mathbb{C}}$. Hence $\mathfrak{s}$ is semisimple. ∎

For example, let $\mathfrak{k} = \mathfrak{u}(n)$. The form $-\operatorname{tr}(XY)$ satisfies both identities in (6.1) without averaging (Exercise 6.1). The proof of the splitting proposition, Proposition 6.2, uses only these identities, so it applies to this form. Here $\mathfrak{z} = i\mathbb{R}I$ and $\mathfrak{s} = \mathfrak{su}(n)$. The splitting of $X \in \mathfrak{u}(n)$ is

$$X = \tfrac{1}{n}(\operatorname{tr} X)\, I + \Big(X - \tfrac{1}{n}(\operatorname{tr} X)\, I\Big) \in i\mathbb{R}I \oplus \mathfrak{su}(n).$$

6.2 Maximal tori and their conjugacy

Definition 6.3. A *torus* is a compact connected abelian Lie group. A *maximal torus* of K is a torus subgroup $T \subseteq K$ not properly contained in another torus subgroup of K.

For example, the diagonal unitary matrices are a maximal torus of $\mathrm{U}(n)$, and the circle $T = \{k_\theta\}$ of Lecture 3 is a maximal torus of $\mathrm{SU}(2)$. § 6.5 proves this.

Let T be a torus with Lie algebra $\mathfrak{t}$. For $t \in T$ and $Y \in \mathfrak{t}$, differentiating $t \exp(sY) t^{-1} = \exp(sY)$ at $s = 0$ gives $\operatorname{Ad}(t)Y = Y$. Putting $t = \exp(rZ)$ with $Z \in \mathfrak{t}$ and differentiating at $r = 0$ gives $[Z, Y] = 0$. So $\mathfrak{t}$ is abelian, and $\exp(X + Y) = \exp X \exp Y$ on $\mathfrak{t}$ (Lecture 2).

Thus $\exp(\mathfrak{t})$ is a subgroup containing a neighbourhood of e (Lecture 2). Such a subgroup is open. It is also closed, because its complement is a union of open cosets. As T is connected, $\exp\colon \mathfrak{t} \to T$ is surjective.

We cite two further facts (H §11.1; S Ch. 5), and Lecture 7 proves the first:

$$T \cong \mathfrak{t}/\Lambda \text{ with } \Lambda = \ker \exp \text{ a lattice}; \qquad \overline{\exp(\mathbb{R}H)} = T \text{ for some } H \in \mathfrak{t}. \tag{6.2}$$

The second is Kronecker's theorem. We call such an H *generic*. Kronecker's theorem also gives $t_0 \in T$ whose powers are dense in T. For example, on the two-torus T^2 of Lecture 3 the irrational line L_α is dense, by the irrational-line proposition. So a nonzero element of $\mathfrak{t}$ tangent to L_α is generic.

Lemma 6.4. *A torus $T \subseteq K$ is maximal if and only if $\mathfrak{t}$ is a maximal abelian subalgebra of $\mathfrak{k}$. Maximal tori exist.*

Proof. Let $\mathfrak{t}$ be maximal abelian, and let $T \subseteq T'$ with T' a torus. Then $\mathfrak{t} \subseteq \mathfrak{t}'$ with $\mathfrak{t}'$ abelian, so $\mathfrak{t} = \mathfrak{t}'$. Hence $T' = \exp(\mathfrak{t}') = \exp(\mathfrak{t}) = T$.

Conversely, let T be maximal, and let $\mathfrak{a}$ be an abelian subalgebra with $\mathfrak{t} \subseteq \mathfrak{a}$. Since $\mathfrak{a}$ is abelian, $\exp(\mathfrak{a})$ is a subgroup, as for $\mathfrak{t}$ above. This subgroup is connected and abelian. Its closure A is a Lie group by the closed subgroup theorem (Lecture 3). The group A is compact, connected and abelian, so it is a torus containing T. Since T is maximal, $A = T$. Hence $\mathfrak{a} \subseteq \mathrm{Lie}(A) = \mathfrak{t}$.

For existence, take a maximal abelian subalgebra $\mathfrak{a}$. As in the second paragraph, $A = \overline{\exp(\mathfrak{a})}$ is a torus. Its Lie algebra is abelian and contains $\mathfrak{a}$, so it equals $\mathfrak{a}$. By the first paragraph, A is a maximal torus. ∎

Theorem 6.5 (Torus theorem; BD §IV.1; S Ch. 5; H §§11.2–11.5)**.** *Let K be compact and connected, and let T be a maximal torus.*

(a) Every $g \in K$ lies in xTx^{-1} for some $x \in K$.

(b) Every maximal torus of K equals xTx^{-1} for some $x \in K$.

Take $K = \mathrm{U}(n)$ with T the diagonal matrices. Then part (a) is the spectral theorem for unitary matrices (Self-check 2). Part (b) says that a maximal torus is diagonal after a change of orthonormal basis. So the theorem extends diagonalisation to every compact connected group.

Hall (H) proves this for compact matrix Lie groups, and Bröcker–tom Dieck (BD) for general compact connected Lie groups. Part (b), the conjugacy of maximal tori, and a Lie algebra form of part (a) have a short complete proof.

Proposition 6.6. *Let T be a maximal torus of K. For every $X \in \mathfrak{k}$ there is $x \in K$ with $\mathrm{Ad}(x)X \in \mathfrak{t}$. Moreover, any two maximal tori of K are conjugate.*

We rotate X by $\mathrm{Ad}(K)$ until it is as close as possible to a generic element $H \in \mathfrak{t}$, using the compactness of K. At the best position the first derivative vanishes, and this says that the rotated element commutes with H.

Proof. Choose a generic $H \in \mathfrak{t}$. Define

$$f \colon K \to \mathbb{R}, \qquad f(x) = \langle \mathrm{Ad}(x)X, H \rangle.$$

It is continuous, so it attains its maximum on the compact group K at some x_0. Put $X_0 = \mathrm{Ad}(x_0)X$ and fix $Y \in \mathfrak{k}$. By Lecture 2,

$$f\big(\exp(sY)x_0\big) = \langle \mathrm{Ad}(\exp sY)X_0, H\rangle = \langle e^{s\,\mathrm{ad}\,Y}X_0, H\rangle.$$

Since f is maximal at x_0, this function of s is maximal at $s = 0$. Hence its derivative $\langle [Y, X_0], H\rangle$ at $s = 0$ vanishes. Using the second identity in (6.1) twice, together with antisymmetry of the bracket, we get

$$0 = \langle [Y, X_0], H\rangle = -\langle X_0, [Y, H]\rangle = \langle X_0, [H, Y]\rangle = -\langle [H, X_0], Y\rangle.$$

Since Y is arbitrary, $[H, X_0] = 0$.

Next we pass from the single element H to all of $\mathfrak{t}$. This step uses genericity of H. Consider

$$S = \{t \in T : \mathrm{Ad}(t)X_0 = X_0\}.$$

It is a closed subgroup of T, since $t \mapsto \mathrm{Ad}(t)X_0$ is continuous. From $[H, X_0] = 0$, every term after the first in the exponential series vanishes. So

$$\mathrm{Ad}(\exp sH)X_0 = e^{s\,\mathrm{ad}\,H}X_0 = X_0 \qquad (s \in \mathbb{R}).$$

Thus S contains $\exp(\mathbb{R}H)$, and hence its closure T. Now let $Z \in \mathfrak{t}$. Differentiating $\mathrm{Ad}(\exp sZ)X_0 = X_0$ at $s = 0$ gives $[Z, X_0] = 0$. So $\mathfrak{t} + \mathbb{R}X_0$ is an abelian subalgebra containing $\mathfrak{t}$. By the characterisation of maximal tori in Lemma 6.4, $\mathfrak{t}$ is maximal abelian, so $X_0 \in \mathfrak{t}$.

Now let T' be a maximal torus with generic element H'. Choose x with $\mathrm{Ad}(x)H' \in \mathfrak{t}$. Then

$$x\exp(sH')x^{-1} = \exp\big(s\,\mathrm{Ad}(x)H'\big) \in T \qquad (s \in \mathbb{R}).$$

Conjugation by x is a homeomorphism and T is closed. Taking closures, $xT'x^{-1} \subseteq T$. The left side is a maximal torus, so equality holds. ∎

Definition 6.7. Let $T \subseteq K$ be a maximal torus. Its *normaliser* is $N_K(T) = \{n \in K : nTn^{-1} = T\}$. The *Weyl group* is $W(K, T) = N_K(T)/T$.

The group $W(K, T)$ acts on T by $nT \cdot t = ntn^{-1}$ and on $\mathfrak{t}$ by $\mathrm{Ad}(n)$. For $n \in N_K(T)$ and $H \in \mathfrak{t}$, the curve $s \mapsto n\exp(sH)n^{-1}$ lies in T. Differentiating it at $s = 0$ gives $\mathrm{Ad}(n)H \in \mathfrak{t}$. Since $\mathrm{Ad}(n)$ is invertible and $\mathfrak{t}$ is finite-dimensional, $\mathrm{Ad}(n)$ maps $\mathfrak{t}$ onto itself. These actions are well defined on cosets because T is abelian.

Lemma 6.8. *Let T be a maximal torus of K and $N = N_K(T)$. Then the identity component N_0 equals T, and $W(K, T)$ is finite.*

Proof. The group N is closed in K, so it is a compact Lie group (Lecture 3). Let $n \in N$ and $H \in \mathfrak{t}$. Then

$$n\exp(sH)n^{-1} = \exp\big(s\,\mathrm{Ad}(n)H\big) \in T \qquad (s \in \mathbb{R}).$$

Differentiating at $s = 0$ gives $\mathrm{Ad}(n)H \in \mathfrak{t}$. If $\exp H = e$, the display gives $\exp(\mathrm{Ad}(n)H) = e$. So $A_n = \mathrm{Ad}(n)|_{\mathfrak{t}}$ maps the lattice Λ into itself.

First, N_0 centralises T. Choose a basis $\lambda_1, \dots, \lambda_r$ of $\mathfrak{t}$ inside Λ. Each map $n \mapsto A_n\lambda_i$ is continuous with values in the discrete set Λ. Since N_0 is connected, this map is constant on

N_0, equal to its value λ_i at e. So $A_n = I$ for $n \in N_0$. Then $n \exp(H) n^{-1} = \exp H$ for all $H \in \mathfrak{t}$. Since $\exp(\mathfrak{t}) = T$, every $n \in N_0$ commutes with every element of T.

Second, $\mathrm{Lie}(N) = \mathfrak{t}$. Let $X \in \mathrm{Lie}(N)$. Then $\exp(sX) \in N_0$ for all s. By the first step, $\mathrm{Ad}(\exp sX)H = H$ for $H \in \mathfrak{t}$. Differentiating at $s = 0$ gives $[X, H] = 0$. So $\mathfrak{t} + \mathbb{R}X$ is abelian, and $X \in \mathfrak{t}$ because $\mathfrak{t}$ is maximal abelian (Lemma 6.4). The connected group N_0 is generated by $\exp(\mathrm{Lie}\, N)$ (Lecture 2). Hence N_0 is generated by $\exp(\mathfrak{t}) = T$, so $N_0 = T$.

Finally, N_0 is open in N (Lecture 2). The cosets of N_0 form a cover of the compact group N by disjoint open sets. Hence there are finitely many cosets. Since $N_0 = T$, the group $W(K, T) = N/N_0$ is finite. ∎

6.3 An outline of the torus theorem via mapping degree

Since $x \exp(X) x^{-1} = \exp(\mathrm{Ad}(x)X)$, the conjugacy statement of Proposition 6.6 shows that every element of $\exp(\mathfrak{k})$ lies in a conjugate of T. Conversely, every conjugate of an element of $T = \exp(\mathfrak{t})$ is an exponential. So part (a) of the torus theorem, Theorem 6.5(a), is equivalent to surjectivity of exp. Surjectivity of exp needs a global argument. We outline the proof in Hall (H §§11.3–11.5).

For connected K define

$$\Phi \colon K/T \times T \to K, \qquad \Phi(xT, t) = xtx^{-1}.$$

This is well defined because T is abelian. The source $K/T \times T$ and the target K are compact connected manifolds of dimension $\dim K$ (Lecture 3).

We count how many times Φ covers a point, with signs. A point y is a *regular value* of a smooth map if the differential of the map is invertible at every point of the inverse image of y. For a smooth map between compact connected oriented manifolds of equal dimension, the signed count of preimages is the same at every regular value. A point outside the image would have count zero. So it is enough to show that the count is nonzero at one regular value. On a circle, an orientation is a direction of travel. The general definition is part of the cited theory.

Figure 6.1 shows the count on the circle, for the maps $f(x) = x + 0.9 \sin 2x$ and $g(x) = 1.5 + \sin x$. A preimage counts $+1$ where the graph rises and -1 where it falls. The map f has count 1 at both levels drawn, and the map g has count 0 at both.

We now carry out the count.

1. Let $t_0 \in T$ be an element whose powers are dense in T, as after (6.2). Then

$$\Phi^{-1}(t_0) = \{(nT,\ n^{-1} t_0 n) : nT \in W(K, T)\}.$$

   Each listed point maps to t_0. Conversely, let $xtx^{-1} = t_0$. Then $x^{-1} t_0^m x = t^m \in T$ for all $m \in \mathbb{Z}$. Taking closures, $x^{-1} T x \subseteq T$. Both are maximal tori, so equality holds and $x \in N_K(T)$. The set is finite by the finiteness of the Weyl group, Lemma 6.8, which was proved without the torus theorem.

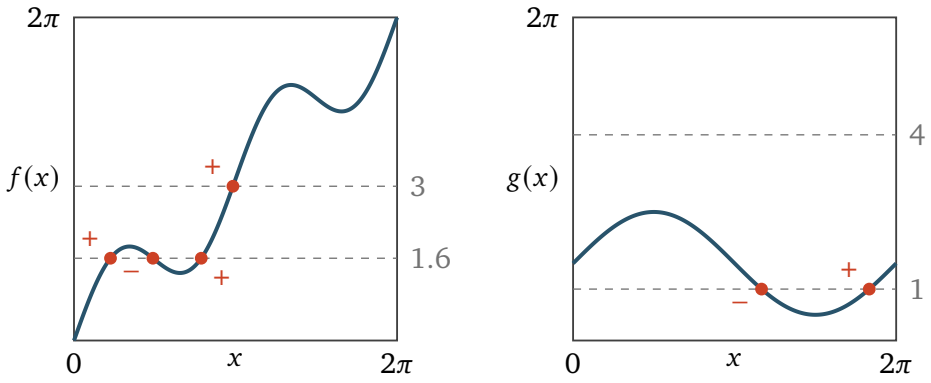


Figure 6.1: Two smooth maps of the circle, drawn on a square whose opposite sides are identified modulo 2π. The dashed lines are levels, numbered at the right, and the marked points are their preimages with the sign of the slope. The map f has signs $+, -, +$ at level 1.6 and $+$ at level 3. The map g has signs $-, +$ at level 1 and no preimage at level 4.

2. The quotient K/T is orientable. Fix orientations of K/T, T and K, and give $K/T \times T$ the product orientation. At each point of $\Phi^{-1}(t_0)$ the differential of Φ is invertible. Its determinant, computed in oriented bases, has the same sign at all of these points.
3. By step 2, t_0 is a regular value of Φ. A smooth map between compact connected oriented manifolds of equal dimension has an integer degree. It is the number of preimages of any regular value, each counted with the sign of the determinant of $d\Phi$ there.
4. So $\deg \Phi = \pm|W(K,T)| \neq 0$. A point outside the image would be a regular value with no preimages, giving degree 0. Hence Φ is surjective.

The omitted step is the degree theory in steps 2–4. It consists of the definition of degree, its independence of the regular value, the orientation of K/T, and the sign comparison at all preimages.

Step 2 rests on a local computation, shown here at (eT, t). By Lecture 3, $Y \mapsto \exp(Y)T$ identifies $\mathfrak{t}^\perp$ with the tangent space of K/T at eT. We identify the tangent spaces of T and of K at t with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation by t. So $Z \in \mathfrak{t}$ corresponds to the curve $s \mapsto t\exp(sZ)$. For $Y \in \mathfrak{t}^\perp$ and $Z \in \mathfrak{t}$,

$$t^{-1}\Phi\big(\exp(sY)T,\ t\exp(sZ)\big) = \exp\big(s\,\mathrm{Ad}(t^{-1})Y\big)\exp(sZ)\exp(-sY).$$

By the product rule, the derivative at $s = 0$ is $\mathrm{Ad}(t^{-1})Y - Y + Z$. The orthogonal map $\mathrm{Ad}(t^{-1})$ fixes $\mathfrak{t}$ pointwise, since T is abelian. Hence it preserves $\mathfrak{t}^\perp$. In these identifications, the differential is block diagonal in $\mathfrak{t}^\perp \oplus \mathfrak{t}$, with blocks $\mathrm{Ad}(t^{-1})|_{\mathfrak{t}^\perp} - I$ and I. Hence

$$\det d\Phi_{(eT,t)} = \det\big(\mathrm{Ad}(t^{-1}) - I\big)\big|_{\mathfrak{t}^\perp}. \tag{6.3}$$

Hall (H §11.5) shows that this determinant is positive at $t = t_0$, and that the sign is the same at the other preimages of t_0. We do not reproduce that part. The same determinant returns in Lecture 9, as the weight in the Weyl integration formula.

6.4 Surjectivity of exp and the Weyl group

Corollary 6.9. *If K is compact and connected, then $\exp\colon \mathfrak{k} \to K$ is surjective.*

Proof. Let $g \in K$. By the torus theorem, Theorem 6.5(a), $g = xtx^{-1}$ with $x \in K$ and $t \in T$. As $\exp\colon \mathfrak{t} \to T$ is surjective, $t = \exp H$ with $H \in \mathfrak{t}$. Hence

$$g = x \exp(H) x^{-1} = \exp\big(\mathrm{Ad}(x)H\big).$$

■

Connectedness is needed. The path $s \mapsto \exp(sX)$ joins e to $\exp X$, so $\exp(\mathfrak{k})$ lies in the identity component. For example, no reflection in $\mathrm{O}(2)$ is an exponential.

The *centraliser* of T is $Z_K(T) = \{g \in K : gt = tg \text{ for all } t \in T\}$. It contains T, since T is abelian.

Theorem 6.10 (H §§11.2–11.5; BD §§IV.1–2). *Let K be compact and connected with maximal torus T. Then $Z_K(T) = T$. Two elements of T are conjugate in K if and only if they lie in one $W(K,T)$-orbit.*

Together with the torus theorem, Theorem 6.5(a), we obtain a bijection

$$T/W(K,T) \longrightarrow \{\text{conjugacy classes of } K\}, \qquad W(K,T)\cdot t \longmapsto \text{class of } t.$$

Surjectivity is the torus theorem. Injectivity is the orbit statement of Theorem 6.10. By Lemma 6.8, the group $W(K,T)$ is finite. So conjugacy in K reduces to the action of a finite group on T.

The Weyl group also has an algebraic description. For $H \in \mathfrak{t}$ the operator $\operatorname{ad} H$ is skew-symmetric for our fixed $\mathrm{Ad}(K)$-invariant inner product. So its complex-linear extension to $\mathfrak{k}_{\mathbb{C}}$ is diagonalisable with imaginary eigenvalues. These operators commute, because $\mathfrak{t}$ is abelian. So they have common eigenvectors. On a common eigenvector, the eigenvalue is i times a real number that depends linearly on H.

A *root* is a nonzero $\alpha \in \mathfrak{t}^*$ such that some nonzero $V \in \mathfrak{k}_{\mathbb{C}}$ satisfies

$$[H, V] = i\alpha(H)V \qquad (H \in \mathfrak{t}).$$

Let R be the set of roots. A root is real on $\mathfrak{t}$. Lectures 7, 8 and 9 keep this convention. The Lie Algebras course writes the same root as the complex-linear form $i\alpha$ on $\mathfrak{t}_{\mathbb{C}}$, and Lecture 7 gives a dictionary. When $\mathfrak{k}$ is semisimple, $\mathfrak{t}_{\mathbb{C}}$ is a Cartan subalgebra of $\mathfrak{k}_{\mathbb{C}}$ in the sense of that course. Lecture 8 proves this.

Define $\alpha^\sharp \in \mathfrak{t}$ by $\langle \alpha^\sharp, H\rangle = \alpha(H)$ for $H \in \mathfrak{t}$. The reflections and the *algebraic Weyl group* are

$$s_\alpha(H) = H - \frac{2\alpha(H)}{\langle \alpha^\sharp, \alpha^\sharp\rangle}\,\alpha^\sharp, \qquad W(R) = \langle s_\alpha : \alpha \in R\rangle \subseteq \mathrm{GL}(\mathfrak{t}).$$

Theorem 6.11 (H §11.7; BD Ch. V). *Let K be compact and connected, with any $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{k}$. Then $nT \mapsto \mathrm{Ad}(n)|_{\mathfrak{t}}$ is an isomorphism from $W(K,T)$ onto $W(R)$.*

6.5 Unitary groups and a noncompact example

Proposition 6.12. *Let $T \subseteq \mathrm{U}(n)$ be the diagonal unitary matrices. For $\sigma \in S_n$ let $P_\sigma e_j = e_{\sigma(j)}$. Then T is a maximal torus,*

$$N_{\mathrm{U}(n)}(T) = \{DP_\sigma : D \in T,\ \sigma \in S_n\},$$

and $DP_\sigma T \mapsto \sigma$ is an isomorphism $W(\mathrm{U}(n), T) \to S_n$. The action on T is

$$P_\sigma \operatorname{diag}(z_1, \dots, z_n) P_\sigma^{-1} = \operatorname{diag}\left(z_{\sigma^{-1}(1)}, \dots, z_{\sigma^{-1}(n)}\right). \tag{6.4}$$

Proof. The group $T \cong \mathrm{U}(1)^n$ is a torus. Fix $t_* = \operatorname{diag}(c_1, \dots, c_n) \in T$ with distinct entries. A matrix commuting with t_* preserves each eigenline $\mathbb{C}e_j$, so it is diagonal. Hence $Z_{\mathrm{U}(n)}(T) = T$. A torus $T' \supseteq T$ is abelian, so $T' \subseteq Z_{\mathrm{U}(n)}(T) = T$. Thus T is maximal.

Formula (6.4) holds for every $t = \operatorname{diag}(z_1, \dots, z_n) \in T$, because

$$P_\sigma t P_\sigma^{-1} e_{\sigma(j)} = P_\sigma t e_j = z_j e_{\sigma(j)}.$$

So each DP_σ normalises T. Conversely, let $g \in N_{\mathrm{U}(n)}(T)$ and $t' = g t_* g^{-1} \in T$. Then

$$t'(g e_j) = g t_* e_j = c_j\, g e_j.$$

The diagonal matrix t' has the distinct eigenvalues $c_1, \dots, c_n$. So its c_j-eigenspace is a coordinate line $\mathbb{C}e_{\sigma(j)}$. Distinct eigenvalues have distinct eigenlines, so σ is a permutation. Thus $g e_j = d_{\sigma(j)} e_{\sigma(j)}$ with $|d_{\sigma(j)}| = 1$. That is, $g = DP_\sigma$ with $D = \operatorname{diag}(d_1, \dots, d_n)$. Since $P_\sigma P_\tau = P_{\sigma\tau}$ and $P_\sigma D P_\sigma^{-1} \in T$, the map $DP_\sigma T \mapsto \sigma$ is a homomorphism. It is onto, since $P_\sigma \in \mathrm{U}(n)$. It is injective because $DP_\sigma \in T$ only when $\sigma = \mathrm{id}$. ∎

We now compare $W(\mathrm{U}(n), T)$ with the algebraic Weyl group, using the Ad-invariant form $-\operatorname{tr}(XY)$. On $\mathfrak{t} = \{i \operatorname{diag}(a_1, \dots, a_n) : a_j \in \mathbb{R}\}$ this form equals $\sum_j a_j b_j$. Here $\mathfrak{t}_{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$. Let E_{jk} be the matrix unit with 1 in row j and column k. For $H = i \operatorname{diag}(a)$ and $j \neq k$,

$$[H, E_{jk}] = i(a_j - a_k) E_{jk}.$$

Let ε_j be the linear form on $\mathfrak{t}$ with $\varepsilon_j\big(i \operatorname{diag}(a_1, \dots, a_n)\big) = a_j$. The roots are $\alpha_{jk} = \varepsilon_j - \varepsilon_k$ for $j \neq k$, with $\alpha_{jk}^\sharp = i(E_{jj} - E_{kk})$ and $\langle \alpha_{jk}^\sharp, \alpha_{jk}^\sharp \rangle = 2$. Then

$$s_{\alpha_{jk}}\big(i \operatorname{diag}(a)\big) = i \operatorname{diag}(a) - (a_j - a_k)\, i(E_{jj} - E_{kk}).$$

This reflection swaps a_j and a_k. By the permutation formula (6.4), the reflection equals $\operatorname{Ad}(P_{(jk)})|_{\mathfrak{t}}$. Transpositions generate S_n. So $W(R)$ is the image of $W(\mathrm{U}(n), T)$, as the algebraic description of the Weyl group in Theorem 6.11 predicts. Hence conjugacy classes of $\mathrm{U}(n)$ correspond to T/S_n. In other words, a unitary matrix is determined up to conjugacy by its eigenvalues with multiplicity.

Corollary 6.13. *Let $n \geq 2$ and $T_1 = T \cap \mathrm{SU}(n)$. Then T_1 is a maximal torus of $\mathrm{SU}(n)$. The normaliser $N_{\mathrm{SU}(n)}(T_1)$ consists of the DP_σ with $D \in T$ and $\det(DP_\sigma) = 1$, and $DP_\sigma T_1 \mapsto \sigma$ is an isomorphism $W(\mathrm{SU}(n), T_1) \to S_n$.*

Proof. The map $(z_1, \dots, z_{n-1}) \mapsto \operatorname{diag}(z_1, \dots, z_{n-1}, (z_1 \cdots z_{n-1})^{-1})$ shows $T_1 \cong \mathrm{U}(1)^{n-1}$. Let $0 < \epsilon < \pi/(n-1)$. The entries $c_j = e^{i\epsilon(2j-n-1)}$ are distinct, and their product is 1 since $\sum_j (2j-n-1) = 0$. The proof of Proposition 6.12, the normaliser computation for $\mathrm{U}(n)$, applies with this t_*. It shows that T_1 is maximal. It also shows that the normaliser consists of the DP_σ with

$$\det(DP_\sigma) = \det D \cdot \operatorname{sgn}\sigma = 1.$$

Every σ occurs, with $D = \operatorname{diag}(\operatorname{sgn}\sigma, 1, \dots, 1)$. The injectivity argument is unchanged. ∎

The roots match again. The Lie algebra $\mathfrak{t}_1$ of T_1 consists of the $i\operatorname{diag}(a)$ with $\sum_j a_j = 0$. Since $\mathfrak{su}(n)_{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$ is spanned by $\mathfrak{t}_{1,\mathbb{C}}$ and the E_{jk}, the roots are the restrictions of the α_{jk}. They are nonzero, because $\alpha_{jk}^\sharp = i(E_{jj} - E_{kk})$ lies in $\mathfrak{t}_1$ and $\alpha_{jk}(\alpha_{jk}^\sharp) = 2$. So $s_{\alpha_{jk}}$ still swaps a_j and a_k. Put $n_{(jk)} = \operatorname{diag}(-1, 1, \dots, 1)P_{(jk)}$, which lies in $N_{\mathrm{SU}(n)}(T_1)$. Diagonal matrices act trivially on $\mathfrak{t}$, so $\operatorname{Ad}(n_{(jk)})|_{\mathfrak{t}_1} = s_{\alpha_{jk}}$. Hence $W(R)$ is the image of $W(\mathrm{SU}(n), T_1) \cong S_n$.

For $\mathrm{SU}(2)$ the transposition is represented by

$$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \operatorname{diag}(-1, 1)P_{(12)}, \qquad w \operatorname{diag}(z, \bar{z}) w^{-1} = \operatorname{diag}(\bar{z}, z).$$

By the bijection between $T_1/W(\mathrm{SU}(2), T_1)$ and conjugacy classes, every conjugacy class of $\mathrm{SU}(2)$ contains exactly one matrix $\operatorname{diag}(e^{i\theta}, e^{-i\theta})$ with $\theta \in [0, \pi]$. In physics terms, choosing a maximal torus of $\mathrm{SU}(2)$ is like choosing a quantisation axis.

Now compare the connected noncompact group $\mathrm{SL}(2, \mathbb{R})$. Its connectedness was shown in Lecture 2 (Exercise 2.4). By the computation in Lecture 2, the element

$$\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$$

is not in $\exp(\mathfrak{sl}(2, \mathbb{R}))$. So the surjectivity statement of Corollary 6.9 fails for this connected noncompact group. The lines $\mathbb{R}\operatorname{diag}(1, -1)$ and $\mathfrak{so}(2)$ are both maximal abelian. Their nonzero elements have real and imaginary eigenvalues respectively, so they are not Ad-conjugate.

The proof of the conjugacy statement in Proposition 6.6 breaks at the invariant inner product of Lemma 6.1. For $H = \operatorname{diag}(1, -1)$ the operator $\operatorname{ad} H$ has eigenvalues $2, 0, -2$, but a skew-symmetric operator has imaginary eigenvalues. Averaging is unavailable because $\mathrm{SL}(2, \mathbb{R})$ is not compact. Exercise 6.4 gives the details.

6.6 Finite central quotients

The next lecture uses the following global form of the splitting.

Theorem 6.14 (BD §V.8). *Let K be compact and connected. Let $T_0 = Z(K)_0$ be the identity component of the centre. It is a torus with Lie algebra $\mathfrak{z}(\mathfrak{k})$. Let K_{sc} be the simply connected Lie group with Lie algebra $[\mathfrak{k}, \mathfrak{k}]$ (Lecture 4). Then K_{sc} is compact, and the inclusion $[\mathfrak{k}, \mathfrak{k}] \to \mathfrak{k}$ integrates to a homomorphism $\varphi \colon K_{\mathrm{sc}} \to K$. The map*

$$\mu \colon T_0 \times K_{\mathrm{sc}} \to K, \qquad \mu(z, y) = z\,\varphi(y),$$

is a surjective homomorphism with finite central kernel F. Hence $K \cong (T_0 \times K_{\mathrm{sc}})/F$.

Compactness of K_{sc} is Weyl's theorem and is part of the citation.

For example, for $K = \mathrm{U}(n)$ we have $T_0 = \{zI : |z| = 1\}$ and $K_{\mathrm{sc}} = \mathrm{SU}(n)$, which is simply connected (H Ch. 13). Here $\mu(z, A) = zA$. If $zA = I$, then $A = z^{-1}I$ and $\det A = z^{-n} = 1$. Hence

$$F = \{(\zeta, \zeta^{-1}I) : \zeta^n = 1\} \cong \mathbb{Z}/n, \qquad \mathrm{U}(n) \cong \big(\mathrm{U}(1) \times \mathrm{SU}(n)\big)/F.$$

Exercise 6.5 checks surjectivity for $n = 2$.

Self-checks

1. Split
$$X = \begin{pmatrix} 3i & 1 \\ -1 & i \end{pmatrix} \in \mathfrak{u}(2)$$
along $\mathfrak{z} \oplus \mathfrak{su}(2)$.
Answer. The centre component is $\frac{1}{2}(\operatorname{tr} X)I = 2iI$. The remainder is skew-Hermitian with trace zero:
$$X - 2iI = \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}.$$
2. Let T be the diagonal torus of $\mathrm{U}(n)$. Without using the torus theorem, prove that every element of $\mathrm{U}(n)$ lies in a conjugate of T, as in Theorem 6.5(a).
Answer. A unitary U is normal. The spectral theorem for normal matrices gives a unitary V such that $V^{-1}UV$ is diagonal. The matrix $V^{-1}UV$ is also unitary, so it lies in T. Hence $U \in VTV^{-1}$.
3. Are $\operatorname{diag}(i, -i)$ and $\operatorname{diag}(-i, i)$ conjugate in $\mathrm{SU}(2)$? Are $\operatorname{diag}(i, -i)$ and I?
Answer. Yes. With w as above, $w \operatorname{diag}(i, -i)w^{-1} = \operatorname{diag}(-i, i)$. The second pair is not conjugate. Conjugate matrices have equal traces, but the traces of $\operatorname{diag}(i, -i)$ and I are 0 and 2.

Summary

An averaged inner product gives the central and semisimple splitting. Maximal tori are conjugate. For connected K, the cited global torus theorem makes exp surjective. Together with the cited orbit theorem for $W(K, T)$, it identifies the conjugacy classes of K with $T/W(K, T)$. The Weyl group is finite. For $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ it is isomorphic to S_n. Compactness is essential, as the connected group $\mathrm{SL}(2, \mathbb{R})$ shows.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, §§11.1–11.5 and 11.7.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1–3 and V.8.
- S: Sepanski, *Compact Lie Groups*, Chapter 5.